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A Novel Electrochemiluminscence Sensor for the Detection of Butyl Hydroxy Anisd and Gallic Acid Based on Graphene Quantum Dots

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Abstract. Graphene quantum dots (GQDs), with stable photoluminescence (PL) and electrocatalytic activity, has potential application in sensor field. Herein, a facile novel electrochemiluminescence (ECL) sensor based on GQDs and glassy carbon electrode (GCE) was developed to detect Butyl hydroxy anisd (BHA) and Gallic acid. The optimization of ECL detection conditions was investigated. The results showed that the sensor exhibited relatively strong luminescence intensity even without the modification of the GCE, which made this sensor system easy to implement. The developed sensor exhibited a linear response ranging from 5 to 160 μM under the optimum conditions. The method displayed the advantages of high sensitivity, good selectivity, wide linear range, low detection limit and fine reproducibility. Moreover, the practicability of the ECL sensing platform in real water samples showed the satisfactory results, indicating a great potential in the field of pharmaceutical analysis.

Introduction

Nowadays, the food safety issue had attracted great attention. Among plenty of food additives, we focused on two of them, butyl hydroxy anisd (BHA) and Gallic acid. BHA was an effective antioxidant [1]. BHA was mainly used as a preservative and antioxidant in food, food packing, animal feed, petroleum products, elastic, and cosmetics. Since last century, BHA, for its antioxidant performance as it avoided rancidification of food which created unpleasant odors, had been added to fat-containing foods and comestible fats [2]. It had reported from the U.S. National Institutes of Health that BHA based on evidence of carcinogenicity was practically expected to be a human carcinogen, which had occurred in experimental animals [3]. Especially, when executed in high doses as part of their diet, BHA caused cell carcinomas of the forestomach and papilloma in Syrian golden hamsters and rats [4]. When investigate human population statistics, the normal low intake levels of BHA showed no significant connotation with an increased danger of cancer [5].

Similarly, in food industry, Gallic acid was a preservative and antioxidant as well [6]. And it's said that Gallic acid was a thin carbonic anhydrase inhibitor according to a study, by changing the properties of alpha-synuclein which was a protein of leading neurodegenerative diseases, Gallic acid influenced the amyloid protein formation [7]. Gallic acid could be classified as a teratogen and preventing fermentation [8]. Consequently, the development of selective, sensitive, fast and reliable ways of monitoring BHA and Gallic acid were actually important in food industry, clinical diagnostics and biotechnology.

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A kind of chemiluminescence phenomenon which was caused by the electrochemical reaction was called electrochemiluminescence (ECL) [9]. When applying voltage to the electrode, some electroactive substances would create excited state luminophore by assimilating and converting electrochemical potential. Those luminophore would emit luminous signal when they jumped to ground state [10]. In most cases, ECL was a chemical reaction based on electrochemical reaction. However, the essence of chemiluminescent signal was the statistic luminescence behaviors of illuminant which were formed form the process of chemical reaction [11]. Thus, as an analytical technology, it had numerous advantages compare to some traditional detection methods. ECL not only retained the advantage of chemiluminescence like wide linear range, high sensitivity and so on, but also had the advance which chemiluminescence could not compare to such as easy controllability, the more stable reagent [12]. Though luminous signal of ECL emitted in a moment, bthe area where illuminant created and existenced was certainly confirmed, which was favor of capture and detection. So, it always presented higher sensitivity of analysis [13]. ECL was not a process of simply mixture between reactant but a chemiluminescence process under the stimulation of electrochemistry signal. As a result, ECL showed a characteristic of extremely controllability. Because the reaction of electrode was a real object, the reaction processes at the surface of electrode presented a good discrimination of space [14]. Compared with other traditional detection techniques to detect BHA, ECL was faster with higher accuracy and lower cost [9].

Graphene was a crystalline allotrope of carbon and it was a 2-dimensional crystal [15]. Its carbon atoms were tightly packed in a regular atomic-scale hexagonal pattern [16]. The graphene quantum dots (GQDs) were becoming an advanced multifunctional material for its unique optical, electronic properties [17], especially for their enormous potentials for biomedical applications [18]. A review on ECL of nanomaterials for bioanalysis believed that nanomaterials biofunctioned with multitudinous biomolecules could offer excellent ECL signal-transduction platforms for constructing novel biosensing devices [19].

Experimental

Reagents and Chemicals. GQDs was purchased from Nanjing XFNANO material technology company. BHA and Gallic acid powder was obtained from Dr. Ehrensorfer (Germany) and store at 4 ^{o}C when it was not in use. A stock solution of BHA and Gallic acid (5 mM) was prepared in Phosphated-buffered solution (PBS). Further, different concentrations of BHA and Gallic acid solutions were diluted step by step with PBS.

PBS (PH=7.4, 0.1 M) was prepared using 0.1 M monopotassium phosphate (KH_2PO_4) and 0.2 M sodium hydroxide (NaOH). 0.1 M Potassium chloride (KCl) and 0.1 M Potassium persulfate ($K_2S_2O_8$), which were purchased from Coway International TechTrans Co., Ltd, were dissolved in PBS which was used as coreactant.

All chemicals and reagent were of analytical reagent grade and used without further purification. All solutions were prepared exclusively in deionize water. Finally all experiments were carried out at room temperature.

Apparatus and Fabrication of ECL Sensor. The platform of the analyze was a MPI-E II model electrochemiluminescence analyzer (Xi'an Remax Electronic Science & Technology Co., Ltd., China), which was used to display the ECL measurements with the voltage changes of the photomultiplier tube (PMT). The function of PMT was transforming ECL emission into electrical signals, and its working voltage was set at 800V in the detection. The ECL detector cell was placed in front of the PMT. The detector cell was made of a microbeaker (high: 35 mm, i.d.: 25 mm) and the potential scan was setting from -0.6 ~ -2.1 V. All experiments were carried out with a conventional three-electrode system with a glassy carbon electrode (GCE) electrode as working electrode, a Ag/AgCl (saturated KCl solution) as reference electrode and a platinum wire as counter electrode.

A GCE (3 mm diameter) was used as a working electrode of the ECL biosensor. Before the measurement, the GCE was sequentially polished with 0.3 and 0.05 μm alumina slurry on chamois

leather, followed by ultrasonic treatment in ethanol and double distilled water respectively to obtain a smooth, clean and mirror-like electrode surface. Then, the GCE was allowed to be dried in air at room temperature.

ECL Measurement Procedures. All electrochemical measurements were carried out on the aforementioned workstation with the three-electrode configuration at room temperature. A potential from -0.6 to -2.1 V was applied to the working electrode and the ECL signal was recorded simultaneously. For the measurements of BHA or Gallic acid, the GCE sensor was immersed in the ECL detection cell. The detection solution was fixed to 10 mL PBS containing 0.1 mg/mL GQDs, 0.09 M KCl, 0.09 M $K_2S_2O_8$. The different concentration of BHA or Gallic acid were added to study the electrochemical response by ECL. ECL signals related to the BHA or Gallic acid concentration could be measured.

The voltage of the photo-multiplier tube (PMT), which was used for transforming ECL emission into electrical signals, set at 800 V and the ECL detector cell was placed in front of the PMT in the process of detection.

Results and Discussion

The Optimization of ECL Detection Conditions. Among the factors affecting the performance of the biosensor, the pH of PBS, scan voltage range and concentration of $K_2S_2O_8$ were important in this experiment. It was found that the biosensor performed an optimum response with the PH up to 7.4. Therefore, the optimum pH of PBS was selected as 7.4 [20]. On the other hand, the intensity of ECL evidently increased with the enhancement of $K_2S_2O_8$ concentration from 0.05 M to 0.09 M and reached to a relatively stable platform after 0.09 M. Thus, the optimal concentration of $K_2S_2O_8$ was 0.09 M [21]. The experimental results showed that the lower low voltage was beneficial to enhance ECL intensity, in other hand, destroy the stability of ECL intensity [22].

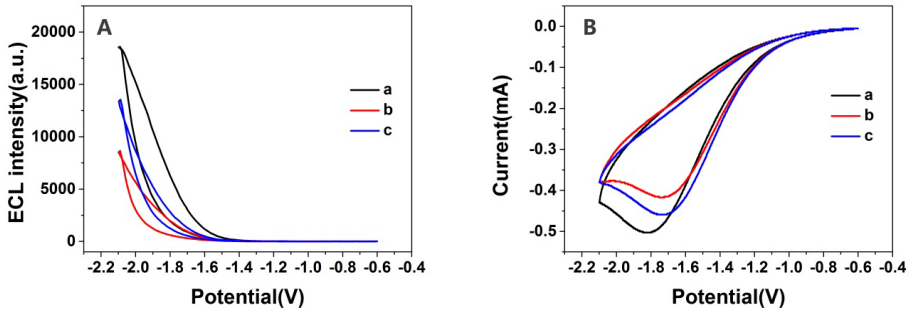


Fig. 1 (A) ECL responses of GQDs: a: without other reagent; b: 160 μM BHA; c:160 μM Gallic acid. (B) CVs of GQDs with GCE electrode a: without other reagent; b:160μM BHA; c:160μM Gallic acid.

Performance of the Biosensor on Detection of BHA and Gallic Acid. In order to demonstate the modification procedures of the GQDs on the electrode, the ECL emission of GQDs was investigated in 10 mL (pH = 7.4) PBS containing 0.1 mg/mL GQDs, 0.09 M KCl and 0.09 M $K_2S_2O_8$ at the range of -0.6 ~ -2.2 V. The ECL responses of the biosensor towards BHA and Gallic acid were also detected in Fig. 1A. It had shown that ECL intensity of GQDs modified electrode in the presence of BHA or Gallic acid was decreased dramatically. The cyclic voltammograms (CVs) of different modified electrodes in 10 mL PBS (pH 7.4) were shown in Fig. 1B. It was observed that the electrochemical current also decreased evidently with the presence of BHA and Gallic acid [23-24]. Compared with ECL responses towards BHA and Gallic acid, the anodic peak was more postively shifted when Gallic acid was added. It indicated that GQDs had more effected on the oxideation of Gallic acid.

Under the optimized conditions, the ECL response of the biosensor towards different concentration Gallic acid was further detected (Fig. 2A). The crresoponding CVs of GQDs with modified electrode was also presented in Fig. 2B. It was illlustated that the ECL signal increased linearly with the increase of Gallic acid concentration in the range of 5~160 μM. It indicated that the Gallic acid could excited the ECL intensity of luminol with GQDs. The calibration curve was presented in the Fig. 2C. And the linear regression equation was

$$Y_1\,(a.u.) = -1917.04\,X_1 + 5596.14\ (R^2 = 0.97246) \quad (1)$$

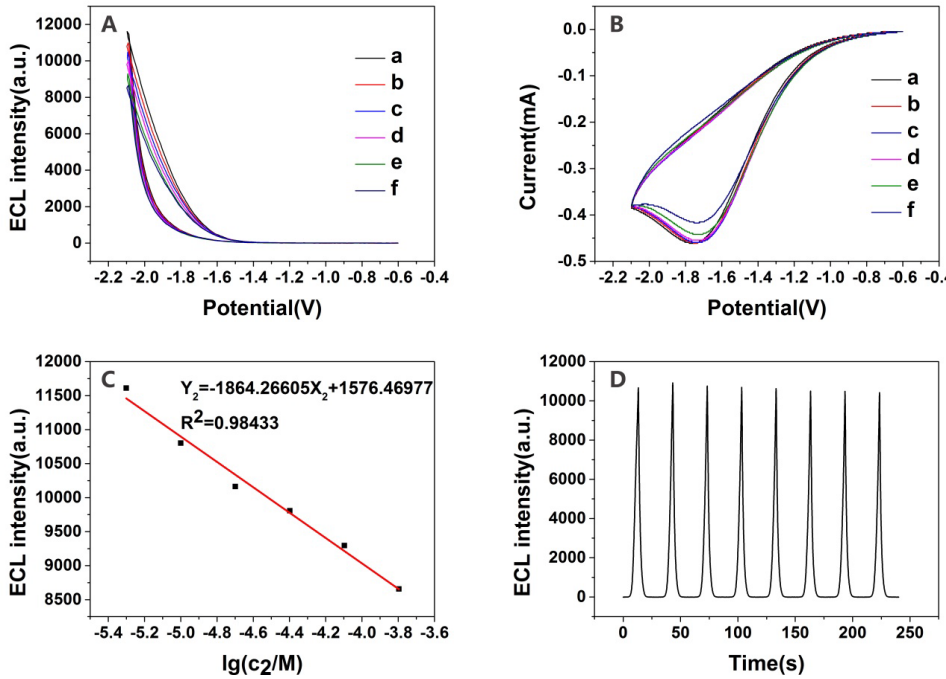


Fig. 2 (A) ECL responses of GQDs with GCE electrode toward different concentration of Gallic acid. a: 5 μM, b: 10 μM, c: 20 μM, d: 40 μM, e: 80 μM, f: 160 μM.
(B) CVs of GQDs with GCE electrode. a: 5 μM, b: 10 μM, c: 20 μM, d: 40 μM, e: 80 μM, f: 160 μM Gallic acid.
(C) Logarithmic calibration curve of the prepared Gallic acid biosensor.
(D) The successively cyclic ECL responses of the biosensor toward 20μM Gallic acid for 8 times.

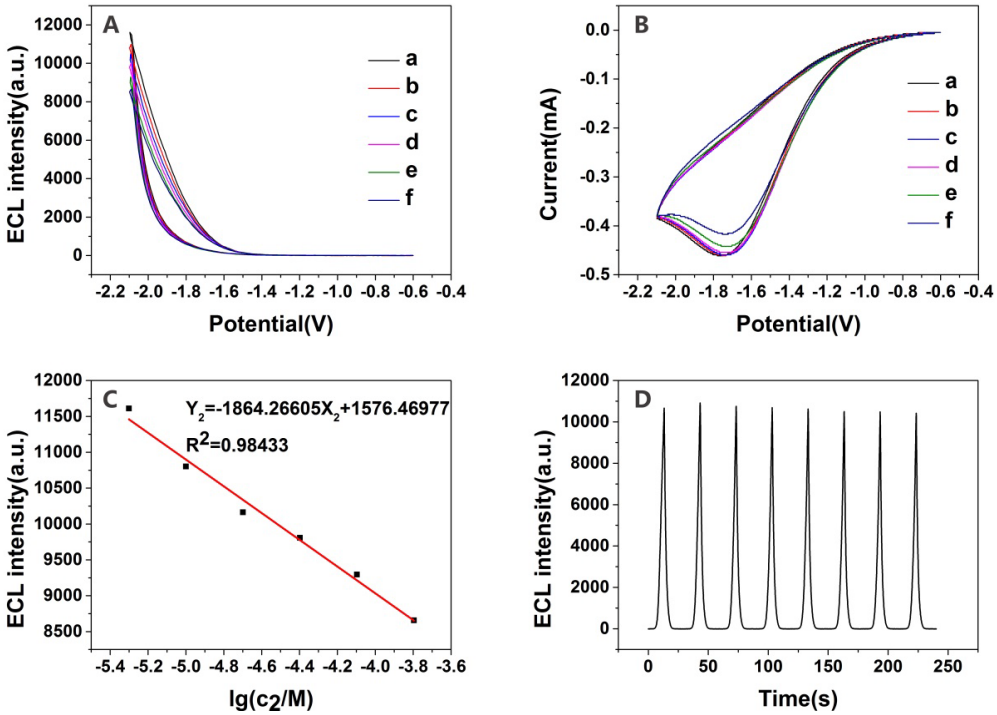


Fig. 3 (A) ECL responses of GQDs with GCE electrode toward different concentration of BHA. a: 5 μM, b: 10 μM, c: 20 μM, d: 40 μM, e: 80 μM, f:160 μM.
(B) CVs of GQDs with GCE electrode. a: 5μM, b: 10μM, c: 20μM, d: 40μM, e: 80μM, f: 160μM Gallic acid.
(C) Logarithmic calibration curve of the prepared BHA biosensor.
(D) The successively cyclic ECL responses of the biosensor toward 20μM Gallic acid for 8 times.

The ECL response of the biosensor towards different concentration BHA and corresponding CVs were also presented in Fig. 3A and 3B. A good linear relationship was observed between the ECL signal and the concentration of BHA in the range of 5~160 μM (Fig. 3C). The linear relapse equation was

$$Y_2\,(a.u.)=-1864.27\,X_2+1576.47\ (R^2=0.98433) \quad (2)$$

In the meantime, the measurements in a lower concentration range were performed. The ECL reply in such a low concentration array was enough to support the rather high sensitivity of the sensor towards Gallic acid and BHA compared with the consequences of most former reports (Table 1 and Table 2)[25-31]. Compared with the results of most previous reports, our biosensor in this work exhibited a higher sensitivity and wider dynamic response range. This might be due to the GQDs with high conductivity for enhanced electrocatalytic response, which benifitered the electron transfer.

Table 1 Comparison of different methods for the determination of BHA.

Method	Linear range/(μM)	Detection limit /(μM)	Reference
AuNPs/ERGO/GCE	0.56 – 55.6	0.23	[25]
Nickel aquapentacyanoferrate	0.61–219.4	0.05	[26]
electrochemiluminescence	5 - 160		This work

Table 2 Comparison of different methods for the determination of Gallic acid.

Method	Linear range /(μM)	Detection limit /(μM)	Reference
CNT-modified electrode	6.25 – 33.7	0.2	[27]
Photoelectrochemical platform	4.17 – 250	1.72	[28]
TiO_2 nanoparticle-modified electrode	5.00 – 150	0.94	[29]
Polyepinephrine/glassy carbon	1.00 – 20	0.663	[30]
Thionine/NiHCF bimediator modified electrode	4.99 – 1200	1.66	[31]
electrochemiluminescence	5 - 160		This work

The Mechanism of this Sensor. According to the diazotization reaction on the modified electrode surface with GQDs, it could require faster electron transfer with GQDs to generated strong ECL signal [15-18]. The possible inhibition mechanism of GQDs on ECL response was summarized as the following Eqs. (3) ~ (6). The schematic presentation of the electrochemiluminescence mechanism of GQDs was shown in Fig. 4.

$$S_2O_8^{2-} + e^- \rightarrow SO_4^{2-} + SO_4^-, \quad (3)$$

$$SO_4^- \rightarrow SO_4^{2-} + h^+, \quad (4)$$

$$C + e^- \rightarrow C^-, \quad (5)$$

$$C^- + h^+ \rightarrow C + \text{light}, \quad (6)$$

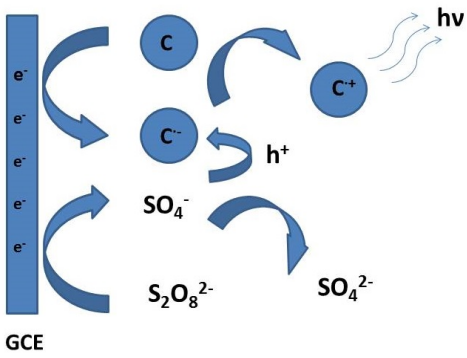


Fig. 4 Schematic presentation of the electrochemiluminescence mechanism of GQDs.

Stability of the Proposed Biosensor. The stability of the mentioned biosensor was studied at the same one biosensor through consecutive scans in PBS including 0.09 M $K_2S_2O_8$, 0.09 M KCl and 0.1 mg/mL GQDs. As shown in Fig. 2D and Fig. 3D, no apparent change in ECL intensity with BHA and Gallic acid was measured after 8 cycles of contiunous measurement. The results demonstrated that the ECL response were kept constantly, indicating an acceptalbe stability of the biosensor.

Conclusion

A novel facile ECL sensor based on GQDs to detect Gallic acid has been developed. The ECL behavior of GQDs has been investigated in detail at the GCE electrode. The ECL signals of GQDs are enhanced by reactant which contains $K_2S_2O_8$ and KCl. In addition, GQDs shows a good linearity for Gallic acid and BHA detection in the range of 5 – 160 μM. Moreover, the novel sensing system has been used to detect BHA and Gallic acid in real water sample with the satisfactory results. The ECL sensor for BHA and Gallic acid detection is high sensitive, selective and convenient in the real analysis application.

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